

MADHAVAN S

FULL-STACK DEVELOPER | AI APPLICATION BUILDER

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SUMMARY

Full-Stack Developer with 5 production-grade projects in Next.js/TypeScript and Django/Python. Focused on multi-tenant data isolation, database concurrency, and applied AI integrations. Track record of dependency security audits, database-level concurrency controls, and validating architecture through load testing.

EDUCATION

B.Sc. Computer Science

2023 – 2026

KMG College of Arts & Science (Thiruvalluvar University), Vellore | CGPA: 7.08/10

TECHNICAL SKILLS

- Languages & Backend: TypeScript, Python, JavaScript (ES6+), Next.js (App Router), tRPC, Prisma, Django REST Framework, Flask, Celery, Redis
- Data & Cloud: Supabase (Postgres, Auth, RLS), Neon PostgreSQL, SQLite, GitHub Actions CI
- Frontend: React 19, Tailwind CSS, Radix UI / shadcn
- AI & Vision: Google Gemini (via Genkit), Groq (LLaMA), face-api.js, Prompt Engineering
- Testing & Security: Vitest, Django TransactionTestCase, k6 load testing, CVE remediation, hmac constant-time comparison, Row-Level Security

WORK EXPERIENCE

Nextgen Technology Pvt. Ltd. | Python Developer Intern

May 2025 – Jun 2025

Vellore, India

Repo: github.com/Madhavan-dev18/havan-transit

- Built a Django REST bus reservation engine (havan-transit) on Neon PostgreSQL, implementing the full booking lifecycle from route search to confirmation.
- Used Django's `select_for_update()` to serialize concurrent seat-booking transactions at the database level, and validated it with a k6 test firing 50 concurrent virtual users at the same seat with zero double-bookings recorded.
- Closed a timing-attack vector in internal token comparison by switching to `hmac.compare_digest()` for constant-time evaluation.
- Offloaded PDF ticket generation to an async Celery + Redis pipeline to keep report rendering off the request/response cycle.
- Load-tested the booking API with k6 at 50 concurrent virtual users: identified a p95 latency of 7.17s exceeding the test's own threshold, and traced it to the hosting tier rather than the locking logic.

ENGINEERING PROJECTS

CHR System — Multi-Tenant Clinical Health Records Platform

2026

Live: chr-system-web.vercel.app | Repo: github.com/Madhavan-dev18/chr-system

Tech: Next.js 15, TypeScript, tRPC v11, Prisma, NextAuth v5, PostgreSQL, Gemini AI

- Designed a HIPAA-aligned EHR monorepo with tenant isolation enforced centrally through a Prisma Client Extension keyed off Node's `AsyncLocalStorage`, rather than repeated per-query checks.
- Added a CI regression guard that greps for a previously-exploited tenant-isolation bypass sentinel on every build, turning a fixed cross-tenant data leak into a permanent, automated check.
- Implemented role-based access control, encrypted record storage, and an AI clinical decision support module with PHI scrubbing before any data reaches the model.

Vivasayi — AI Farm Management Platform

2025 – Present

Built for Smart India Hackathon (college-level round) | Live: vivasayi-wheat.vercel.app | Repo: github.com/Madhavan-dev18/vivasayi

Tech: Next.js 15, TypeScript, React 19, Supabase, Google Gemini via Genkit, Vitest

- Built a production-deployed farm management platform with AI-backed crop diagnosis, advisory, and recordkeeping tools, securing all user data with Supabase row-level security.
- Remediated a critical Next.js remote code execution vulnerability (GHSA-9qr9-h5gf-34mp) and a jsPDF CVE chain through a full dependency audit and enforced secure versioning.
- Built retry logic for Gemini API calls that distinguishes transient errors (which get exponential-backoff retries) from daily quota exhaustion (which skips straight to a fallback model instead of retrying a guaranteed failure).
- Built a 23-language i18n system (React context + custom hook) with full translation coverage across major Indian languages including Tamil and Hindi.

CERTIFICATIONS & LANGUAGES

- Certifications: Python & OOP, Data Science & Analytics, Agile Methodology (Infosys Springboard)
- Languages: Tamil (Native), English (Professional)